

Data and Code Disclosure for

“Supply Chain Risk: Changes in Supplier Composition and Vertical Integration”

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This Readme file is part of the replication package for “Supply Chain Risk: Changes in Supplier Composition and Vertical Integration” to fulfill the data and code disclosure requirement for accepted manuscripts in *Journal of International Economics*. This document provides a roadmap to the data and code used in the above-referenced paper and consists of three main parts. Part I provides a list of the statistical software and packages used and descriptions of the data and code. Part II provides the data dictionaries for all the variables used in the analysis, including both raw and user-generated variables. Part III lists and explains the codes that analyze the cleaned data and generate tables and figures used in the paper. For any questions regarding the code, please reach out to us via email.

I. Statistical software, data, and code

- Statistical software:
 - Stata 14 MP is used for the empirical analysis. The Stata packages listed below are used in the analysis.
 - reghdfe
 - ivreghdfe
 - outreg2
 - Eventus Version 8.0, subscribed through Wharton Research Data Services (WRDS), is used to generate cumulative abnormal returns.
 - R 4.2.0 is used to generate the word clouds. The R packages listed below are used in the analysis.
 - stm
 - wordcloud

- Python 3.10 is used to generate the supply chain risk measure. We install Python and the required packages via a Conda environment. The Python packages listed below are used in the analysis.
 - bs4
 - pip
 - nltk
 - pandas
 - html5lib
 - pdfplumber
- Data files:
 - scrisk.dta: This dataset contains the firm-year level supply chain risk (SCRisk) and supply chain sentiment (SCSentiment) measures we constructed. Section 3.1 of the paper describes the measures in detail. Firms are identified by GVKEY (Global Company Key - a unique identifier and primary key for each company in the Compustat dataset).
 - Compu.dta: This is simulated data that mimics the structure of Compustat data, which is provided by S&P Global Market Intelligence. To gain access to this dataset, please visit WRDS at <https://wrds-www.wharton.upenn.edu/>. Due to data confidentiality agreement, we are not allowed to post the data. This synthetic dataset is provided for the purpose of demonstrating the codes listed in the next sections.
 - customer_supplier_year_revere.dta: This is simulated data that mimics the structure of Factset Revere data, which we use to obtain information on supply chains. Factset Revere collects relationship information from primary public sources, such as SEC 10-K annual filings, investor presentations, and press releases, and classifies the relationship types. To gain access to this dataset, please contact WRDS. Due to data confidentiality agreement with Factset, we are not allowed to post the data. This synthetic dataset is provided for the purpose of demonstrating the codes listed in the next sections.

- sdc.dta: This is simulated data that mimics the structure of the Securities Data Company (SDC) Mergers and Acquisitions database for M&As. We download SDC data from the SDC Platinum terminal. To obtain access to the data, please contact Refinitiv at <https://www.refinitiv.com/en/products/sdc-platinum-financial-securities>. We merge SDC with the Bureau of Economic Analysis's (BEA) Input-Output tables to identify whether the target and the acquirer are in vertically related industries.
- CAR.dta: This is simulated data that mimics the output from Eventus. We use Eventus to calculate cumulative abnormal returns surrounding each M&A announcement in the sample. To gain access to Eventus, please contact WRDS.
- Output_clean.dta: This is the output dataset by applying data cleaning code _1_Code_To_Prepere.do described below. This is also the dataset that generates the main results.
- Code files:
 - _1_Code_To_Prepere.do: This code prepares the raw datasets and creates the variables needed for analysis. In addition, it merges all the datasets to generate the final panel dataset Output_clean.dta for most of the main analysis.
 - _2_Code_for_Regressions_Main.do: This is the main code file for the analysis, which generates Tables 1, 2, 3, and 4.
 - _3_Code_for_return.do: This code prepares the dataset needed to study cumulative announcement returns and performs the analysis reported in Table 5.
 - Figure1.R: This code generates the word cloud reported in Figure 1.

II. Data dictionary

Dataset: scrisk.dta

Variable name	Definition
gvkey	Firm identifier
year	Year

SupplyChainUncertainty	Firm-level supply chain risk measure constructed from earnings calls
SupplyChainSentiment	Firm-level supply chain sentiment measure constructed from earnings calls

Dataset: Compu.dta

Variable name	Definition
year	Year
gvkey	Firm identifier
sic2	2-digit SIC industry code
multinational	An indicator variable that equals one if the firm has geographical segments in other countries based on Compustat segment data
major	An indicator variable that equals one if the firm reports major customers that purchase at least 10% of its sales in Compustat segment data
fc_ww	An indicator variable that equals one if the firm is financially constrained according to Whited and Wu (2006) measure
fc_hp	An indicator variable that equals one if the firm is financially constrained according to Hadlock and Pierce (2010) measure
at	Total assets
ceq	Total common equity
csho	Number of common shares outstanding
prcc_f	Closing price (fiscal year end)
sale	Total sales
ib	Income before extraordinary items
che	Cash and short-term investments
instown_perc	Fraction of shares owned by financial institutions

Dataset: customer_supplier_year_revere.dta

Variable name	Definition
gvkey_customer	Customer identifier
ind_customer	Customer industry code
continent_customer	Customer continent
country_customer	Customer country
year	Year
at_customer	Customer total assets
gvkey_supplier	Supplier identifier
scrisk_supplier	Supplier SCRisk
continent_supplier	Supplier continent

country_supplier	Supplier country
ind_supplier	Supplier industry code
at_supplier	Supplier total assets
sale_supplier	Supplier total sales
supplier_above_med3	Supplier sales is above the 3-digit SIC industry median

Dataset: sdc.dta

Variable name	Definition
year	Year
gvkey_acquiror	Acquiror identifier
gvkey_target	Target identifier
pct_input	Percentage of input that is used by the acquiror industry from the target industry
pct_output	Percentage of output that is made by the acquiror industry to the target industry
acquiror_sic	Acquiror industry code
target_sic	Target industry code

Dataset: CAR.dta

Variable name	Definition
year	Year
gvkey	Firm identifier
car11	Cumulative abnormal return (CAR) over a three-day event window [-1, +1] around an M&A announcement, obtained by estimating the market model over a [-255, -31] day estimation window

Dataset: Output_clean.dta

Variable name	Definition
year	Year
gvkey	Firm identifier
ma_input_d	A dummy variable that equals one if the firm conducts an M&A with a firm from an upstream industry according to the Bureau of Economic Analysis's (BEA) Input-Output tables
ma_output_d	A dummy variable that equals one if the firm conducts an M&A with a firm from a downstream industry according to the Bureau of Economic Analysis's (BEA) Input-Output tables
ma_nonrelated_d	A dummy variable that equals one if the firm conducts an M&A with a firm that is not in an upstream, downstream, or the same industry

re_size	Focal firm's total assets scaled by its suppliers' average total assets
firm2_scriskIV2	The maximum SCRisk of a firm's suppliers
sic2	2-digit SIC industry code
multinational	An indicator variable that equals one if the firm has geographical segments in other countries based on Compustat segment data
major	An indicator variable that equals one if the firm reports major customers that purchase at least 10% of its sales in Compustat segment data
fc_ww	An indicator variable that equals one if the firm is financially constrained according to Whited and Wu (2006) measure
fc_hp	An indicator variable that equals one if the firm is financially constrained according to Hadlock and Pierce (2010) measure
profit1	Income before extraordinary items scaled by total assets
cash	Cash and cash equivalent securities scaled by total assets
SupplyChainUncertainty	Firm-level supply chain risk measure constructed from earnings calls
SupplyChainSentiment	Firm-level supply chain sentiment measure constructed from earnings calls
lag_ln_at	Natural logarithm of total assets, lagged measure at t-1
lag_tobinq1	Assets minus cash and cash equivalent securities plus book value on equity scaled by assets, lagged measure at t-1
ln_at	Natural logarithm of total assets
lag_cash	Cash and cash equivalent securities scaled by total assets, lagged measure at t-1
lag_profit1	Income before extraordinary items scaled by total assets, lagged measure at t-1
lag_firm2_scriskIV2	The maximum SCRisk of a firm's suppliers, lagged measure at t-1
num_supplier	A firm's total number of suppliers
number_supplier_same_cont	The total number of suppliers in the same continent as the firm
number_supplier_above_med3	A firm's total number of suppliers with sales above the median of their 3-digit SIC industry
number_supplier_us	A firm's total number of suppliers in the U.S.
number_supplier_diff_cont2	The total number of suppliers in different continents as the firm
mean_supplier_ind	The average of a firm's number of suppliers by input industry
diff_cont_frac	Fraction of a firm's suppliers located in a continent different from that of the firm over the total number of suppliers
m_share_sic2_sale	Firm's sales scaled by the 2-digit SIC industry's total sales

instown_perc	Fraction of shares owned by financial institutions
tobinq1	Assets minus cash and cash equivalent securities plus book value on equity scaled by assets
lag_SCUncertainty	Firm-level supply chain risk measure constructed from earnings calls, lagged measure at t-1
lag_SCSentiment	Firm-level supply chain sentiment measure constructed from earnings calls, lagged measure at t-1

III. Analysis

Below is the list of tables and figures in the paper and the codes that produce them. Note that if you run the codes on the simulated data or output files generated by the simulated data, the results will not match what is reported in the paper, as those are not the actual data.

- Figure 1: Figure1.R
- Table 1: _2_Code_for_Regressions_Main.do
- Table 2: _2_Code_for_Regressions_Main.do
- Table 3: _2_Code_for_Regressions_Main.do
- Table 4: _2_Code_for_Regressions_Main.do
- Table 5: _3_Code_for_return.do